

```
from google.colab import files
uploaded = files.upload()
```

[Choose files](#) Screenshot ... 143144.png

Screenshot 2026-07-20 143144.png(image/png) - 133941 bytes, last modified: 20/07/2026 - 100% done
Saving Screenshot 2026-07-20 143144.png to Screenshot 2026-07-20 143144 (2).png

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Use a recently uploaded image path
image_path = 'Screenshot 2026-07-20 143144 (2).png'

img = cv2.imread(image_path)

if img is None:
    print(f"Error: Could not load image from {image_path}. Please make sure the file exists and is ac
else:
    rows, cols = img.shape[:2]

    # Define source and destination points for Homography
    # src_points are the corners of the original image
    src_points = np.float32([[0, 0], [cols - 1, 0], [0, rows - 1], [cols - 1, rows - 1]])

    # dst_points define how the original corners will be mapped, creating a perspective effect
    # This creates a trapezoidal effect where the bottom of the image is narrowed
    dst_points = np.float32([[0, 0], [cols - 1, 0], [0, int(0.7 * rows)], [cols - 1, int(0.7 * rows)]])

    # Find the Homography matrix
    M, _ = cv2.findHomography(src_points, dst_points)

    # Apply the perspective transformation using the homography matrix
    homography_img = cv2.warpPerspective(img, M, (cols, rows))

    # Convert images from BGR to RGB for correct display with matplotlib
```

```
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
homography_img_rgb = cv2.cvtColor(homography_img, cv2.COLOR_BGR2RGB)

# Display the original and transformed images using matplotlib
plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(homography_img_rgb)
plt.title('Homography Transformed Image')
plt.axis('off')

plt.show()

# If you want to save the transformed image, you can uncomment the line below:
# cv2.imwrite('transformation_using_Homography_Image.jpg', homography_img)
```

Original Image



Homography Transformed Image



